SaralAI

सरल AI · Voice-First Government Services Assistant

Complete Technical & Architecture Reference

Type: Project Documentation · Full Stack Reference

Version: MVP Frontend Complete

Updated: March 2026

Stack: Vite + Vanilla JS · Node.js / FastAPI (planned)

Status: Frontend ✅ Complete · Backend ❌ Planned

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SaralAI — Complete Project Document

Voice-First Government Services Assistant · Technical & Architecture Reference

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1. Project Identity

Field	Value
Name	SaralAI (सरल AI)
Meaning	"Simple AI" in Hindi
Tagline	"A simple helper to understand government services"
Type	Voice-first AI assistant for government scheme guidance
Stack	Vite + Vanilla JS (frontend) · Node.js/FastAPI (backend — planned)
Run	`cd c:\Users\siddh\Desktop\SaralAI && npm run dev` → http://localhost:5173

What SaralAI IS vs IS NOT

✅ SaralAI IS	❌ SaralAI IS NOT
A voice-first **explainer** of government schemes	A scheme discovery engine (myScheme does this)
A **simplifier** of complex government language	A government portal replacement
A **step-by-step guide** for applying	An application submission system
A **complementary layer** on top of myScheme	A ChatGPT/Gemini clone
A **client-side, stateless guidance** tool	An eligibility decision maker

2. What Is Done vs Remaining

✅ DONE — Frontend (Current Session)

Phase	What Was Built
Project Foundation	Vite app, `index.html`, `package.json`, `src/styles/index.css` design system
Routing	`src/router.js` — hash-based router for 12 routes
State Management	`src/state.js` — app-wide state with scheme result storage
Icons Library	`src/icons.js` — SVG icon kit (mic, check, arrows, gov, etc.)
Components	All 8 components: Header, Button, MicButton, LanguageCard, AudioPlayer, ProgressIndicator, ChecklistItem, StepCard
All 12 Screens	LandingScreen, LanguageScreen, InteractionModeScreen, SpeakScreen, TypeScreen, ListeningScreen, ProcessingScreen, ExplanationScreen, ClarificationScreen, WhatNextScreen, DocumentsScreen, GuidanceScreen
AI Engine	`src/ai.js` — keyword-based scheme search + intent detection across all 10 MVP schemes
Scheme Data	`src/Schemes.json` — all 10 MVP schemes curated with full details
Data Integration	ProcessingScreen runs real queries → ExplanationScreen shows real scheme data → DocumentsScreen pulls real documents → GuidanceScreen shows real steps
Voice Input	Web Speech API integrated into SpeakScreen + ListeningScreen
Clarification	ClarificationScreen shows top 4 matching scheme cards when query is unclear
WhatNext	Fully dynamic summary with real scheme name, category, and official portal link

❌ REMAINING — Backend & Deployment

Phase	What Needs Building
Backend API	Node.js/FastAPI endpoints for `/api/query`, `/api/scheme`, `/api/intent`
Real STT	Google Cloud Speech or Azure STT for Indian language accuracy
Real TTS	Google TTS or Azure TTS for voice playback of explanations
LLM Integration	Gemini/OpenAI API for advanced simplification prompts
Database	PostgreSQL for scheme data (currently client-side JSON)
PWA	Service worker, offline mode, installable
Deployment	Vercel/Railway frontend, Cloud Run backend
Multi-language UI	Hindi/regional language text in all screens
Analytics	Privacy-safe usage analytics

3. System Architecture

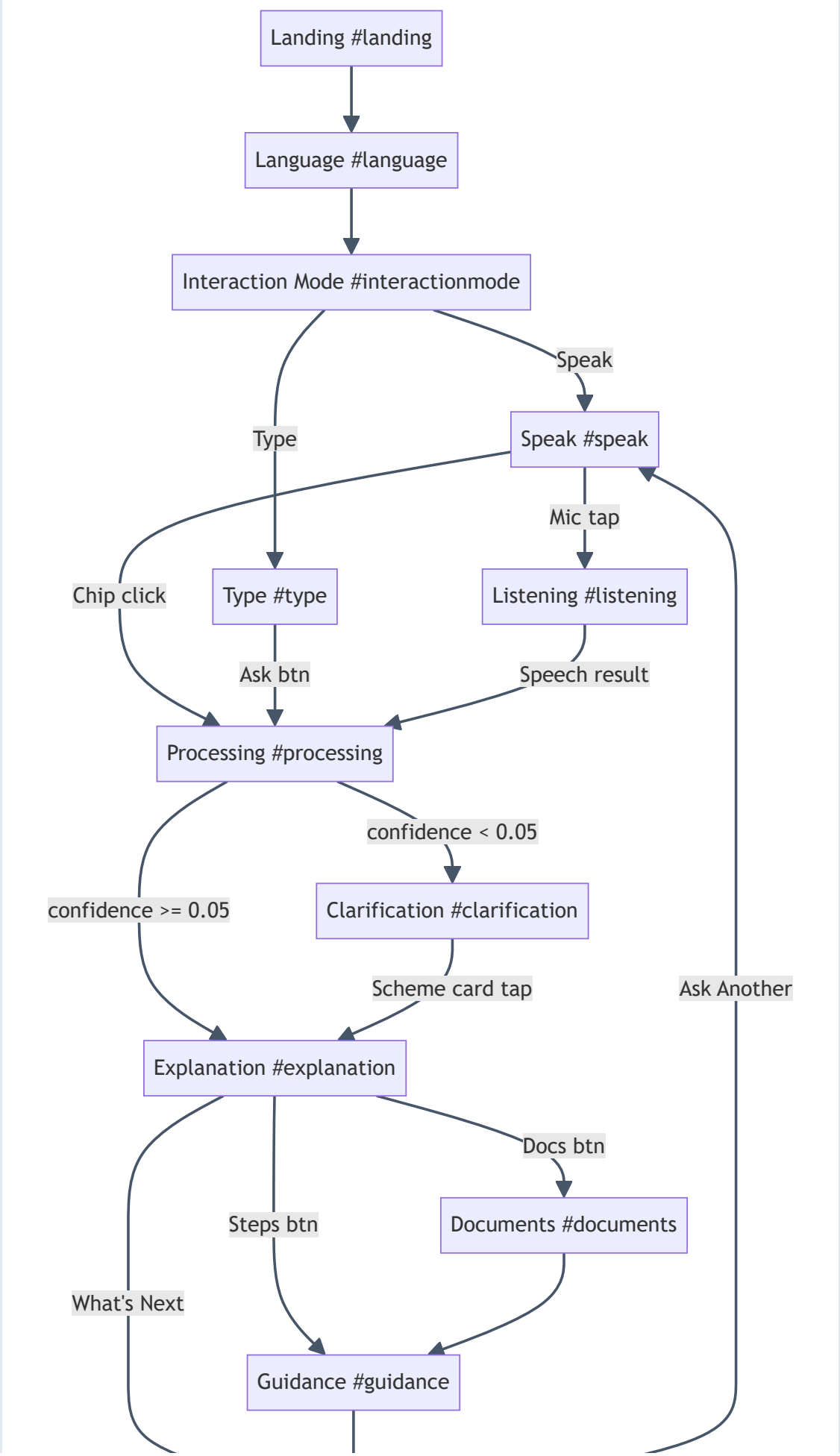
High-Level Flow

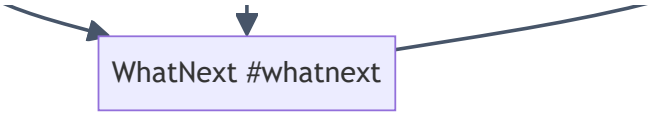
```
User (Voice/Text)
       |
       ▼
┌──────────────────────┐
|    FRONTEND (Vite App)   |
|  http://localhost:5173   |
|                          |
|  ┌──────────────────┐ |
|  |  src/router.js      | |  ← Hash router (#landing, #speak, etc.)
|  |  src/state.js       | |  ← Shared state (query, scheme, language)
|  |  src/ai.js          | |  ← Client-side AI engine
|  |  src/Schemes.json   | |  ← Curated scheme knowledge base
|  |  src/screens/       | |  ← 12 screen modules
|  |  src/components/    | |  ← 8 reusable UI components
|  └──────────────────┘ |
└──────────────────────┘
       |
       ▼ (PLANNED — not yet built)
┌──────────────────────┐
|    BACKEND API           |
|  (Node.js / FastAPI)     |
|                          |
|  POST /api/query         |
|  POST /api/speech-to-text|
|  POST /api/text-to-speech|
|  GET  /api/scheme/:id    |
|  POST /api/intent        |
└──────────────────────┘
       |
       ▼
┌──────────────────────┐
|  EXTERNAL SERVICES       |
|  ├── Google STT API      |
|  ├── Google TTS API      |
|  ├── Gemini/OpenAI LLM   |
|  └── PostgreSQL DB       |
└──────────────────────┘
```

Data Flow (Current — Client-Side)

```
User types query (TypeScreen)
         |
         ▼
setState({ currentQuery })           [src/state.js]
         |
navigate('processing')               [src/router.js]
         |
         ▼
processQuery(query)                  [src/ai.js]
  ├── searchSchemes(query)
  |     └── scores all 10 schemes by keyword overlap
  |         returns ranked [{scheme, confidence}]
  ├── detectIntent(query)
  |     └── classifies as OVERVIEW/ELIGIBILITY/DOCUMENTS/STEPS
  └── generateExplanation(scheme, intent)
        └── returns structured explanation object
         |
         ▼
confidence < 0.05?
  ├── YES → navigate('clarification')   shows top 4 scheme cards
  └── NO  → setState({ currentScheme, currentExplanation, currentIntent })
              navigate('explanation')
                   |
                   ├── Documents btn → navigate('documents')
                   |     pulls state.currentScheme.required_documents
                   └── Steps btn → navigate('guidance')
                         pulls state.currentScheme.application_process.steps
```

Navigation Flow


Landing #landing
Language #language
Interaction Mode #interactionmode
Speak
Type
Speak #speak
Mic tap
Chip click
Type #type
Listening #listening
Ask btn
Speech result
Processing #processing
confidence < 0.05
confidence >= 0.05
Clarification #clarification
Scheme card tap
Ask Another
Explanation #explanation
Docs btn
Steps btn
Documents #documents
What's Next
Guidance #guidance



4. Frontend — Full Breakdown

4.1 Project Setup

`index.html` — Single HTML entry point

```
<div id="app"></div>
<script type="module" src="/src/main.js"></script>
```

- `#app` is the mount point for all screens
- `type="module"` enables ES module imports across all JS files

`package.json` — Minimal deps

```
{ "scripts": { "dev": "vite", "build": "vite build" }, "devDependencies": { "vit
e": "^7.2.4" } }
```

No React, no framework. Pure vanilla JS with Vite for bundling. Vite handles hot module reload, ES module bundling, fast dev server.

`src/main.js` — App entry point (452 lines)

- Imports all 12 screens and 8 components
- Calls `injectStyles()` — collects all CSS exported from screen/component files and appends to
- Calls `initRouter()` — starts hash-based routing
- Defines global `window.SaralAI.showToast()`, `showLoading()`, `hideLoading()` utilities

- Defines shared animation CSS: screen transitions, sound wave animation, skeleton loading, glassmorphism, stagger effects

---

4.2 Design System — `src/styles/index.css`

All CSS variables defined in `:root`. Every value referenced as CSS custom property.

Color Palette (Government Trust)

```
--color-primary: #2563EB        /* Government blue */
--color-primary-dark: #1D4ED8
--color-primary-bg: rgba(37,99,235,0.08)
--color-bg: #F7F8FA             /* Light page background */
--color-bg-card: #FFFFFF        /* Card backgrounds */
--color-text-primary: #1A1A2E
--color-text-secondary: #64748B
--color-text-muted: #94A3B8
--color-success: #10B981
--color-error: #EF4444
```

Typography Scale

```
--font-family: 'Inter', -apple-system, sans-serif
--font-size-xs: 0.75rem  (12px)
--font-size-sm: 0.875rem (14px)
--font-size-base: 1rem   (16px)
--font-size-lg: 1.125rem (18px)
--font-size-xl: 1.25rem  (20px)
--font-size-2xl: 1.5rem  (24px)
--font-size-3xl: 1.875rem (30px)
```

Spacing Scale: `--space-1` (4px) → `--space-20` (80px)

Border Radius: `--radius-sm` (6px) → `--radius-full` (9999px)

Animation keyframes defined: `fadeIn`, `slideUp`, `slideDown`, `scaleIn`, `bounce`, `wave`, `shimmer`

Utility classes: `.container`, `.screen`, `.screen-content`, `.screen-center`, `.card`, `.card-interactive`, `.card-selected`, `.btn`, `.btn-primary/secondary/ghost/dark/lg/sm`, `.icon/icon-sm/icon-lg/icon-xl`, `.animate-fadeIn/slideUp/etc.`

4.3 Routing — `src/router.js`

How it works: Hash-based routing using `window.location.hash`

```
// All 12 routes
const routes = {
  landing, language, interactionmode, speak, type,
  listening, processing, explanation, guidance,
  documents, clarification, whatnext
}

// Navigation
function navigate(route) {
  window.location.hash = route;  // triggers hashchange event
}

// On hashchange → calls onRouteChangeCallback
// main.js registers: onRouteChange(route => renderScreen(route))
// renderScreen adds exit animation, then calls screen.render() + screen.init()
```

Key design decisions:

- Hash router = no server-side routing needed, works on any static host
- Each screen is rendered as an HTML string, not a real DOM tree (no virtual DOM)
- `exit animation (200ms)` → clear `#app.innerHTML` → render new screen string → call `init()` for event bindings

---

4.4 State Management — `src/state.js`

Flat singleton object with pub/sub listeners.

```
const state = {
  // Language
  selectedLanguage: 'en',
  languages: [{code, native, english}],  // 8 languages

  // Interaction
  interactionMode: 'voice',   // 'voice' | 'text'
  isListening: false,
  isProcessing: false,

  // Query
  currentQuery: '',           // User's typed/spoken question

  // Step tracking
  currentStep: 1,
  totalSteps: 3,

  // Query Results (set by ProcessingScreen after ai.js runs)
  currentScheme: null,        // Full scheme object from Schemes.json
  currentIntent: null,        // 'OVERVIEW' | 'ELIGIBILITY' | 'DOCUMENTS' | 'STEP
S'
  currentExplanation: null,   // { schemeName, summary, eligibilityPoints, benefi
tPoints, steps, ... }
  topMatches: [],             // All ranked scheme matches (for clarification scr
een)
  queryHistory: []            // All previous queries this session
}
```

API:

- `getState()` → returns shallow copy of state
- `setState(updates)` → merges updates, notifies all subscribers
- `subscribe(fn)` → registers listener, returns unsubscribe fn
- `getSelectedLanguage()` → returns language object for current code
- `setSelectedLanguage(code)` → convenience wrapper

4.5 AI Query Engine — `src/ai.js`

Purpose: Replaces a backend AI service with a pure client-side JavaScript keyword engine. Reads all data from `Schemes.json` at import time.

`searchSchemes(query)` — Scheme Search

Scores all 10 schemes using keyword overlap:

```
// Each scheme has a keyword list like:
SCHEME_KEYWORDS = {
  PM_KISAN: ['kisan', 'farmer', 'kheti', 'pm kisan', 'samman nidhi', ...],
  PM_JAY: ['ayushman', 'health', 'hospital', 'cashless', 'bimari', ...],
  ...
}

// Scoring algorithm:
function scoreScheme(query, schemeId, scheme) {
  let score = 0, totalWeight = 0;
  // 1. Direct keyword match: score += 1 per keyword hit
  // 2. Scheme name word match: score += 0.5
  // 3. Category match: score += 0.3
  return score / totalWeight;  // 0-1 confidence
}
```

Returns `[{scheme, confidence}]` sorted descending.

`detectIntent(query)` — Intent Classification

Four intent buckets with keyword lists:

```
INTENT_KEYWORDS = {
  DOCUMENTS: ['document', 'papers', 'kya chahiye', 'kaagaz', 'aadhaar', ...],
  STEPS: ['apply', 'kaise', 'process', 'karna', 'fill', 'form', ...],
  ELIGIBILITY: ['eligible', 'kaun', 'paatra', 'milega', 'can i apply', ...],
  OVERVIEW: ['what is', 'kya hai', 'explain', 'batao', 'yojana', ...]
}
```

Counts keyword matches per intent bucket → returns highest scoring intent.

`generateExplanation(scheme, intent)` — Response Generator

Reads from scheme object, builds a structured response:

```
return {
  schemeId, schemeName, category, intent,
  summary,              // context-aware lead sentence
  leadPoints,           // primary bullets based on intent
  eligibilityPoints,    // from scheme.eligibility_criteria[].condition
  benefitPoints,        // from scheme.benefits.details[]
  steps,                // from scheme.application_process.steps[]
  documents,            // from scheme.required_documents[]
  confusions,           // from scheme.common_confusions[]
  limitations,          // from scheme.limitations_and_notes[]
  officialSource,       // from scheme.source_information.official_website
  applicationMode,      // from scheme.application_process.mode
  disclaimer: 'This information is for guidance only.'
}
```

`processQuery(query)` — Main Pipeline

```
export function processQuery(query) {
  const ranked = searchSchemes(query);       // search all 10 schemes
  const { scheme, confidence } = ranked[0]; // top match
  if (confidence < 0.05) return { result: null, confidence, topMatches: ranked };
  const intent = detectIntent(query);        // classify intent
  const result = generateExplanation(scheme, intent);
  return { result, confidence, topMatches: ranked };
}
```

Threshold: confidence < 0.05 → go to Clarification screen. ≥ 0.05 → show Explanation.

4.6 Icons Library — `src/icons.js`

Returns inline SVG strings. Used as `${getIcon('mic', 'icon')}` inside HTML template literals.

Available icons: `mic`, `micOff`, `stop`, `waves`, `arrowRight`, `arrowLeft`, `check`, `x`, `info`, `government`, `keyboard`, `translate`, `idCard`, `language`

Each icon is a pure SVG

with `currentColor` stroke, no external files needed.

4.7 Components (8 Files)

`Header.js` — Sticky top bar

- **Props**: `showNav`, `showLanguageToggle`, `showVoiceAssist`, `showBack`
- Shows SaralAI logo (links to #landing) by default
- `showLanguageToggle=true` → shows "English / English" pill → navigates to #language on click
- `showNav=true` → shows nav links (Home, Services, Language, Help)

- `initHeader()` → attaches click listeners to back and language toggle buttons

`Button.js` — Reusable button

- **Props**: `text`, `variant` (primary/secondary/ghost/dark), `size` (default/lg/sm), `icon`, `iconPosition`, `fullWidth`, `id`, `disabled`
- Returns HTML string: `[]`
- Icons rendered inline from `icons.js`

`MicButton.js` — Microphone button

- `MicButton({size, isListening, id})` → large circle button with pulse animation rings
- `MicButtonSmall({id, text})` → compact mic button with label below
- Animated CSS rings expand outward when `isListening=true`

`LanguageCard.js`

- `LanguageCard({code, native, english, selected})` → single language card
- `LanguageGrid(languages, selectedLanguage)` → 2×4 grid of language cards
- Selected state: blue border + checkmark overlay

`AudioPlayer.js`

- `AudioPlayer({title, subtitle, duration, currentTime})` → "Listen to explanation" card
- Shows waveform animation bars, play/pause button, time display
- Currently static (no real audio — TTS integration planned for backend phase)

`ProgressIndicator.js`

- `ProgressIndicator({current, total, label})` → progress bar + step label
- Used in GuidanceScreen to show "Step 2 of 5"
- CSS animated fill bar

`ChecklistItem.js`

- `ChecklistItem({title, subtitle, icon, checked, id})` → tappable document item
- `Checklist(items)` → stacked list of checklist items
- Checkbox animates to filled blue when checked

- Used in DocumentsScreen for required documents list

`StepCard.js`

- `StepCard({number, title, description, icon, active})` → single numbered step card
- `NumberedSteps(stepsArray)` → renders step list
- Used in legacy ExplanationScreen (now inline numbered list used in rewrite)

4.8 Screens (12 Files)

Each screen file exports three things:

1. `ScreenName()` → returns HTML string (pure function, no DOM access)
1. `initScreenName()` → called after render, attaches event listeners
1. `screenNameStyles` → CSS string injected into by `main.js`

`LandingScreen.js` — Route: `#landing`

What it does: Welcome screen. Builds trust, explains purpose in one line.

Key elements:

- Animated concentric ring icon (government building SVG inside blue circle)
- Heading: "A simple helper to understand government services"
- Subtitle: "Voice-first AI assistant"
- Single "Start" button → navigates to `#language`

No state reads, no data dependencies. Pure UI.

`LanguageScreen.js` — Route: `#language`

What it does: Language selection. Ensures user comfort from first interaction.

Key elements:

- Reads `state.languages` (8 languages: en, hi, bn, te, mr, ta, gu, kn)
- Renders `LanguageGrid` component
- On card click: calls `setSelectedLanguage(code)`, adds checkmark, removes previous selection
- "Continue" button → navigates to `#interactionmode`
- Footer hint with mic icon: "You can also say the name of the language"

State written: `selectedLanguage`

`InteractionModeScreen.js` — Route: `#interactionmode`

What it does: Choose between voice (Speak) or keyboard (Type) input.

Key elements:

- Two large card-buttons: "Speak" (primary, blue gradient) and "Type" (secondary, grey)
- "Speak" card has "Recommended" badge
- Click Speak → setState `interactionMode: 'voice'` → navigate `#speak`
- Click Type → setState `interactionMode: 'text'` → navigate `#type`
- Footer: "You can switch between modes anytime"

Auto-proceeds on selection (no extra continue button).

`SpeakScreen.js` — Route: `#speak`

What it does: Primary voice query entry point.

Key elements:

- Large `MicButton` center-stage with pulsing rings
- Example query: "Ghar ke liye sarkari madad kaise milegi?"
- 4 quick example chips (PM Kisan, PMAY, Ayushman, Ration Card) — clicking any chip sets `currentQuery` directly and navigates to `#processing` (bypasses voice)
- "Type instead" link → `#type`

Voice logic:

```
// Mic tap with Speech API support:
recognition = new SpeechRecognition();
recognition.lang = 'hi-IN,en-IN';
window._saralaiRecognition = recognition; // pass to ListeningScreen
recognition.start();
navigate('listening');

// Without Speech API support:
navigate('type');  // graceful fallback
```

`TypeScreen.js` — Route: `#type`

What it does: Keyboard input fallback/alternative.

Key elements:

- `</code> with 500 char limit counter</li>`